

PART
1

General
academic
English

PART
2

Reading
English
literatures

PART
3

Poster,
presentation
and
manuscript

PART
4

Participate
conference

- ✓ Introduction to Biology
- ✓ Microbiology
- ✓ Botany
- ✓ Zoology
- ✓ Molecular biology
- ✓ Fermentation engineering
- ✓ Gene engineering and genetically modified foods

Purpose:
Efficiently memorize new
words and extend your
vocabulary, get familiar with
scientific sentences.

Microbiology

bacteria [bæk'tɪriə] n. 细菌 bacterium的复数形式
biomass ['baɪəu'mæs] n. 生物量
modern microbiology 现代微生物学
specialty ['speʃəlti] n. 专业, 专长
food science 食品科学
microbiologist [maɪkrəʊbaɪ'ɒləɡɪst] n. 微生物学家
gene expression 基因表达
enzyme activity 酶活性
recombinant DNA technology 重组DNA技术
genetic engineering 遗传工程
indication [ɪndɪ'keɪʃən] n. 指示, 表示, 迹象
virus ['vaɪrəs] n. 病毒, 病原体
bacteriology [bæktɪrɪə'ɒlədʒi] n. 细菌学

mycology [maɪ'kɒlədʒi] n. 真菌学
phycology [faɪ'kɒlədʒi] n. 藻类学
parasitology [pærəsai'tɒlədʒi] n. 寄生虫学
virology [vaɪ'rɒləgi] n. 病毒学
epidemiology [ˌepɪˌdɪmi'ɒlədʒi] n. 流行病学
microbial ecology 微生物生态学
microbial physiology 微生物生理学
microbial genetics 微生物遗传学
microbial molecular biology 微生物分子生物学
pathogen ['pæθədʒən] n. 病原体
pathology [pə'θɒlədʒi] n. 病理学
immunology [ɪmju'nɒlədʒi] n. 免疫学
infectious [ɪn'fekʃəs] adj. 传染的, 有传染性的
bioinformatics [baɪəʊn'fɒmətɪks] n. 生物信息学

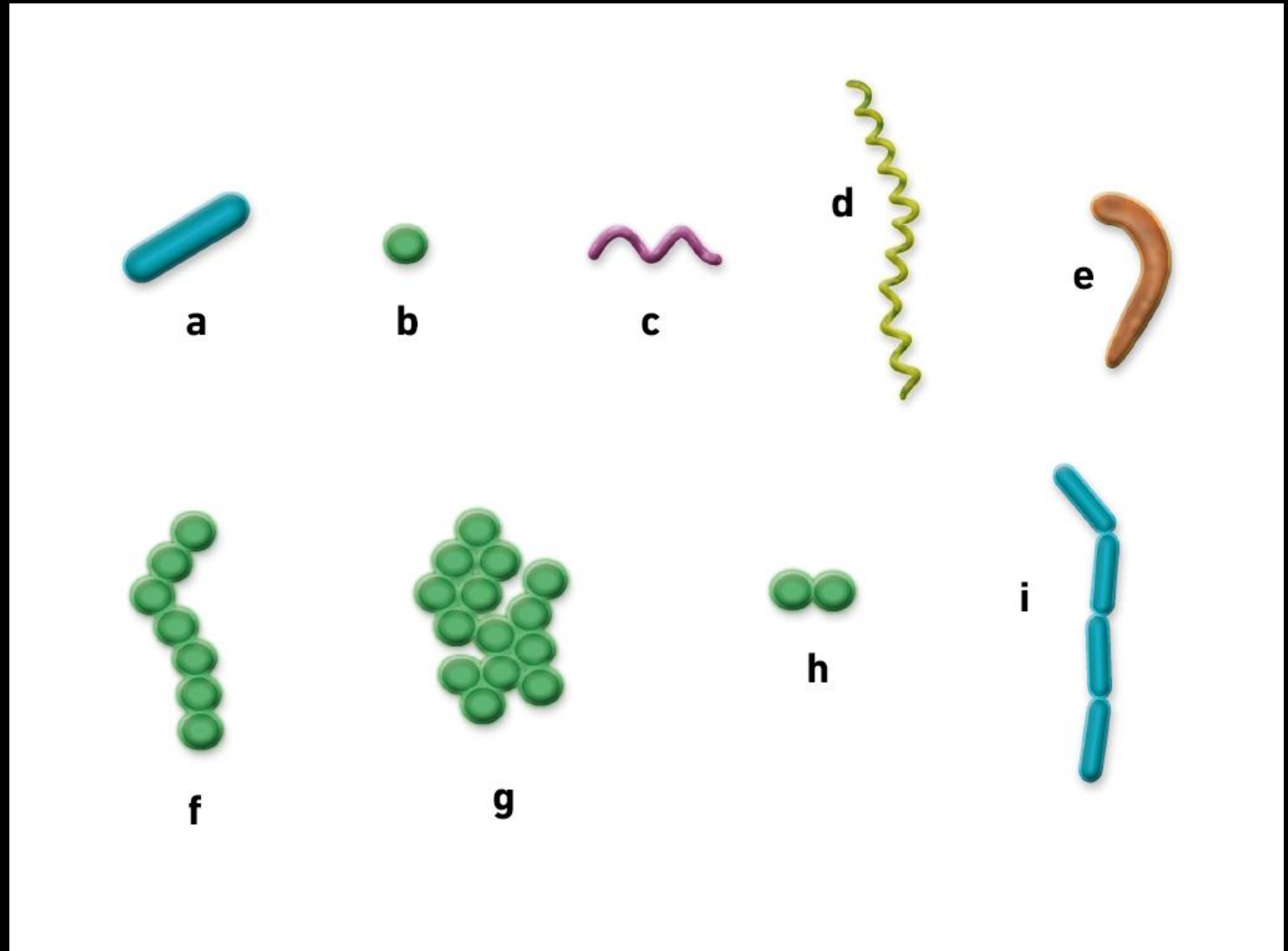
As the scientist-writer Steven Jay Gould emphasized, we live in the Age of Bacteria. They are the first living organisms on our planet, live virtually everywhere, are more numerous than any other kind of organism, and probably constitute the largest component of the earth's biomass. The whole ecosystem depends on their activities, and they influence human society in countless ways. Thus modern microbiology is a large discipline with many different specialties; it has a great impact on fields such as medicine, agricultural and food sciences, ecology, genetics, biochemistry, and molecular biology.

Microbiology is the branch of biology studying microorganisms, divided into many subdisciplines which include bacteriology, mycology, phycology, parasitology, virology, epidemiology, microbial ecology, microbial physiology, microbial genetics and microbial molecular biology. Our need to control infectious diseases has brought about the fields of pathology and immunology. Bioinformatics is a new area of research in microbiology.

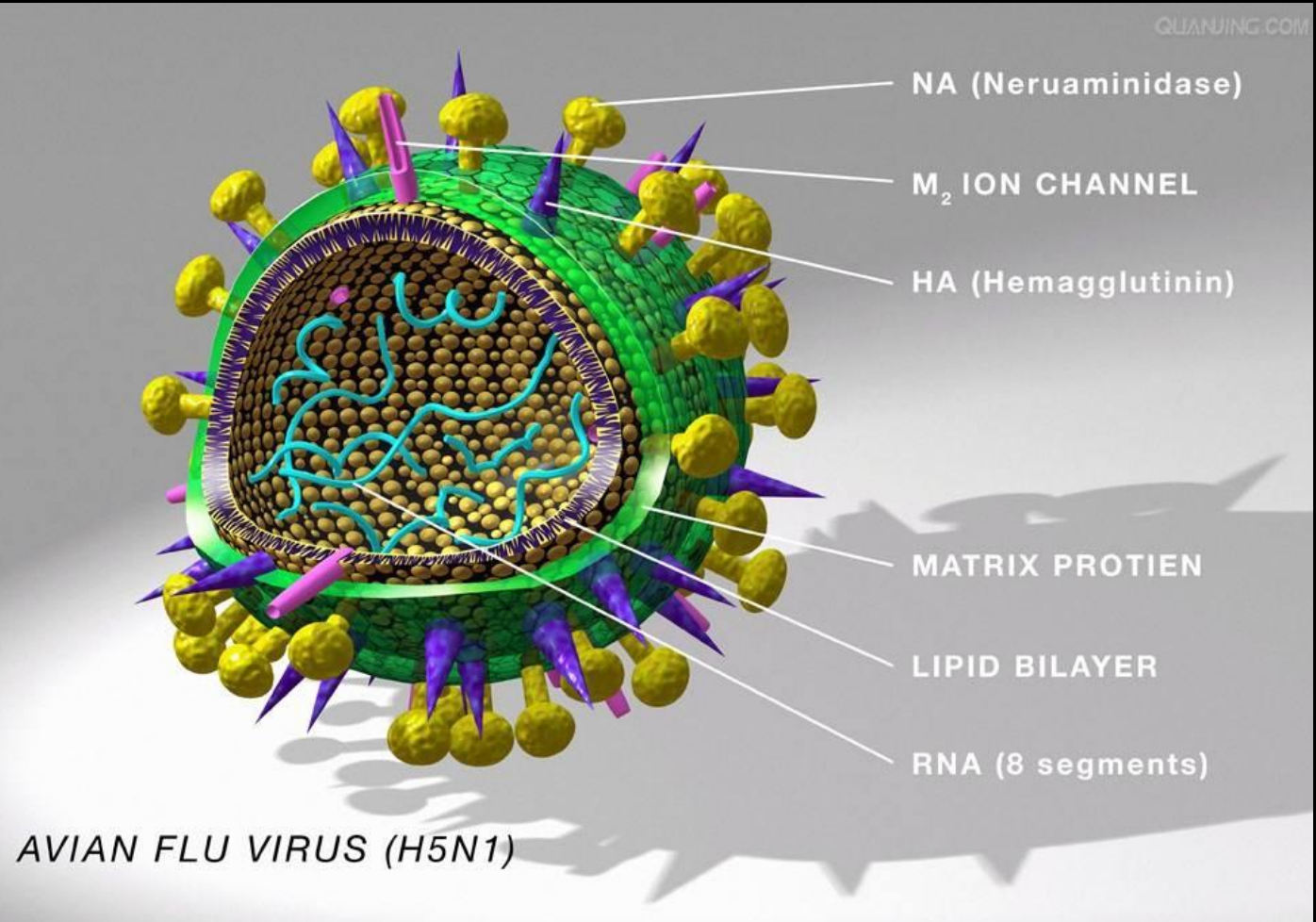
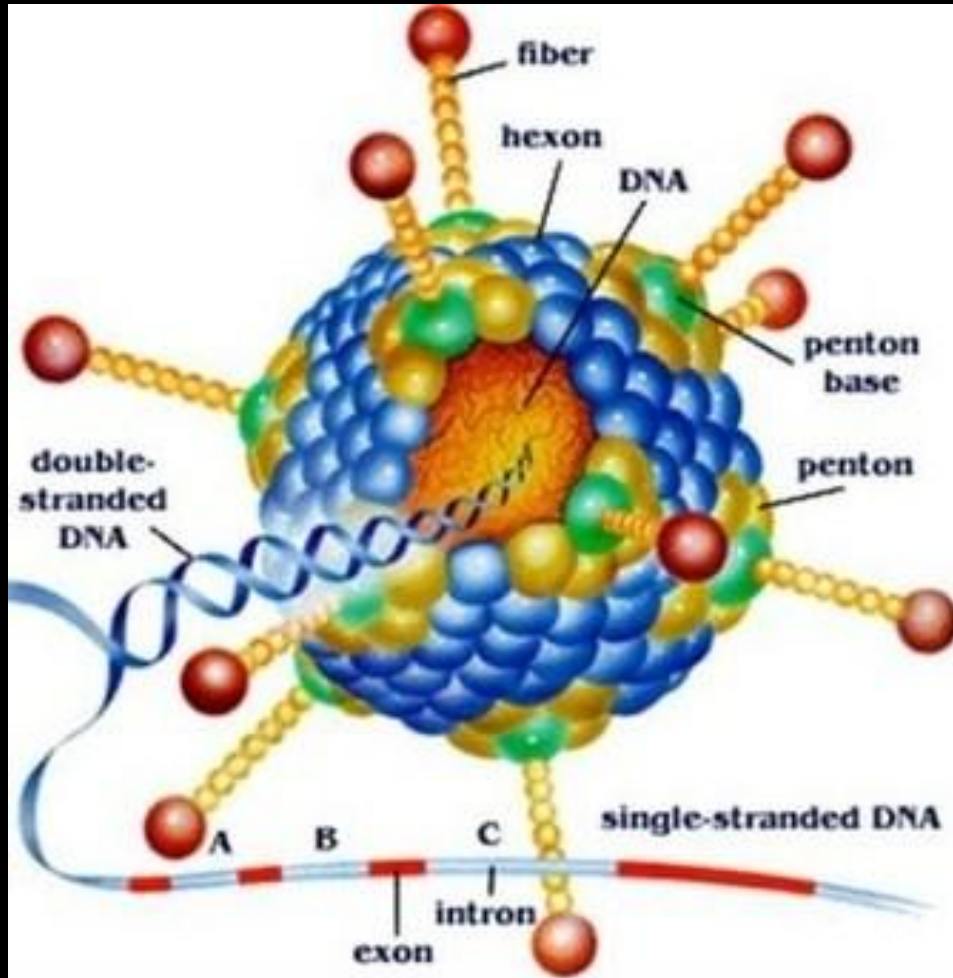
Bacteria

The different bacterial shapes:

- a. bacillus (rod),
- b. coccus (spherical),
- c. spirillum (spiral),
- d. spirochaete (corkscrew),
- e. vibrios (comma),
- f. chain of cocci,
- g. cluster of cocci,
- h. pair of cocci,
- i. chain of bacilli



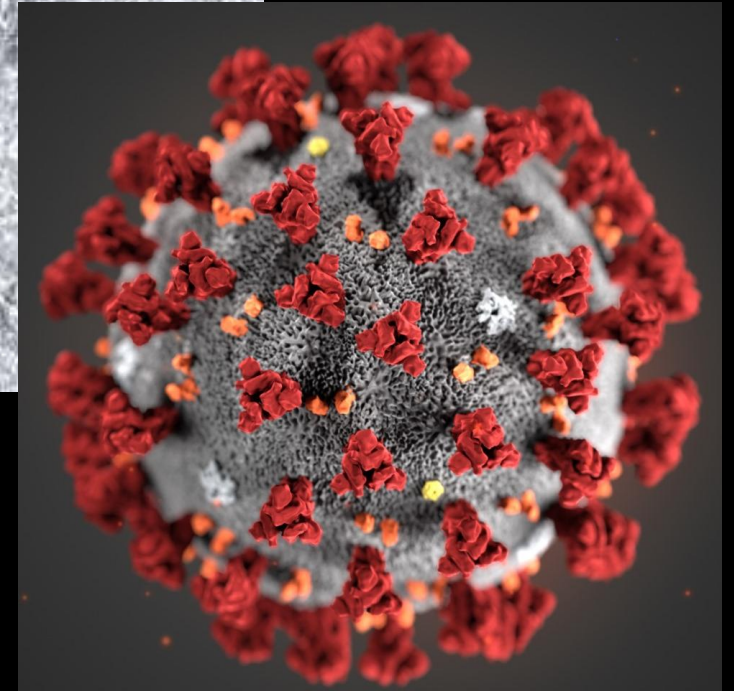
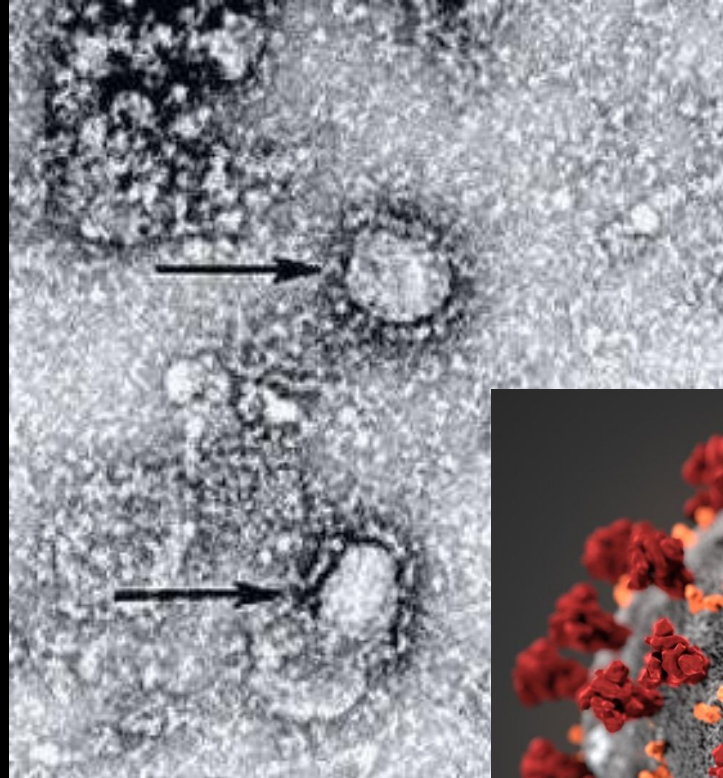
Virus



Virus

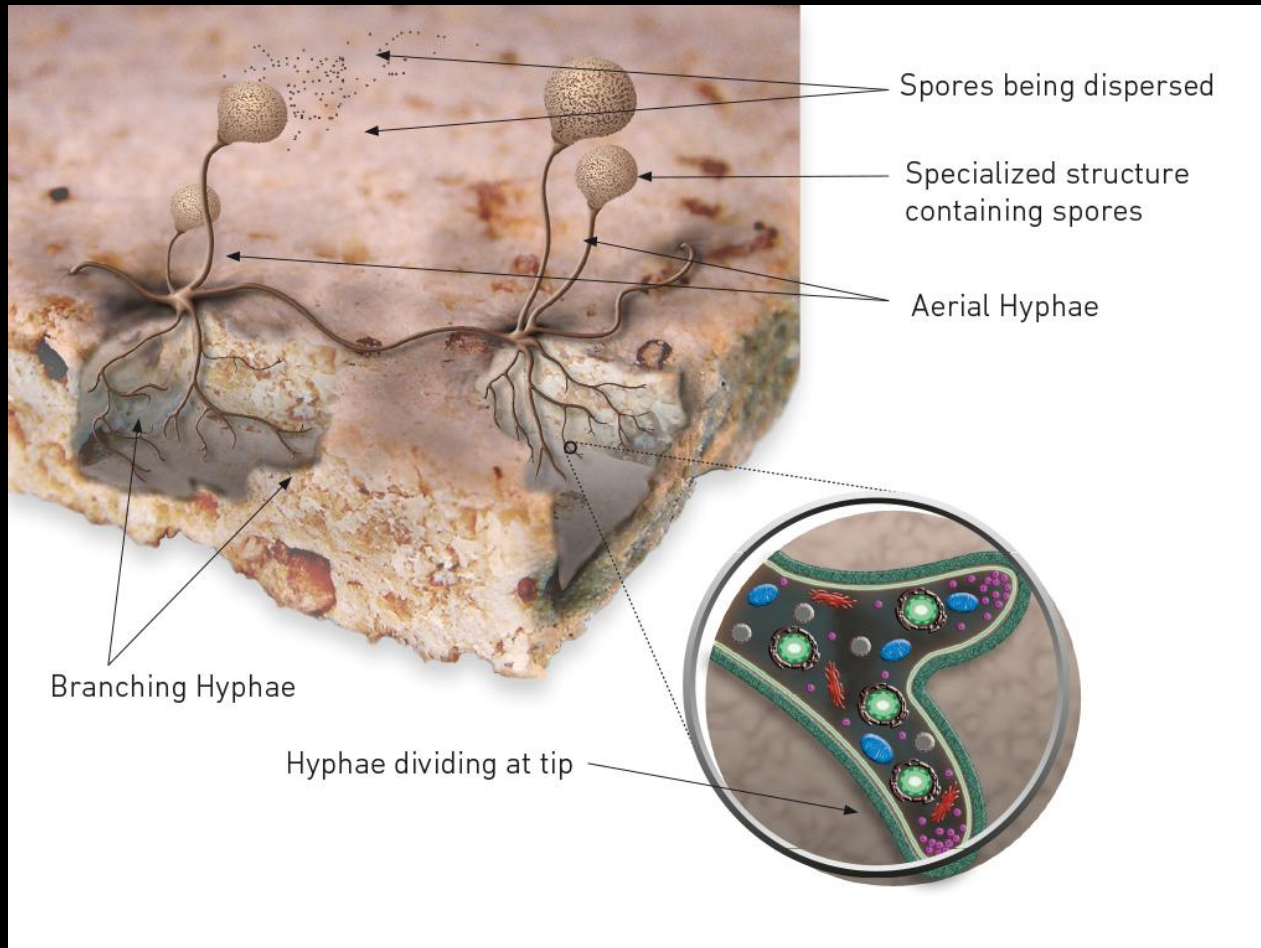


Tobacco mosaic virus which is helical in shape.



SARS-CoV-2, 2019 Novel Coronavirus, nCov

Fungi



How a mycelium is formed and how spores are distributed



infectious disease 传染病

unidentified ['ʌnaɪ'dentɪfaɪd] adj. 未确认的, 无法识别的

creutzfeldt-Jacob disease 克雅病, JC病, 早老性痴呆症

hantavirus ['hæntə'veɪrəs] n. 汉滩病毒

AIDS [eɪdz] abbr. 获得性免疫功能丧失综合症(Acquired Immune Deficiency Syndrome)

Public health microbiologist 公共健康微生物学家

immune system 免疫系统

infectious agents 传染源, 传染物

monoclonal antibody 单克隆抗体

allergy ['ælədʒi] n. 过敏症

autoimmune disease 自身免疫病

rheumatoid arthritis 风湿性关节炎

What are some of the current occupations of professional microbiologists?

One of the most active and important is medical microbiology, which deals with the diseases of humans and animals. Medical microbiologists identify the agent causing an infectious disease and plan measures to eliminate it. Frequently they are involved in tracking down new, unidentified pathogens such as the agent that causes variant creutzfeldt-Jacob disease, the hantavirus, and the virus responsible for AIDS. These microbiologists also study the ways in which microorganisms cause disease.

Public health microbiology is closely related to medical microbiology. Public health microbiologists try to control the spread of communicable diseases. They often monitor community food establishments and water supplies in an attempt to keep them safe and free from infectious disease agents.

Immunology is concerned with how the immune system protects the body from pathogens and the response of infectious agents. It is one of the fastest growing areas in science; for example, techniques for the production and use of monoclonal antibodies have developed extremely rapidly. Immunology also deals with practical health problems such as the nature and treatment of allergies and autoimmune diseases like rheumatoid arthritis.

免疫学关注的是免疫系统怎样保护机体免受病原体侵害以及如何响应传染源。它是科学中发展最快的领域；例如，单克隆抗体的制备及使用技术已经取得了极快速的发展。免疫学也负责解决诸如过敏症和像风湿性关节炎这样的自身免疫疾病的本质和治疗等实际健康问题。

agricultural microbiology 农业微生物学

ruminant ['ru:minənt] n. 反刍动物

bacterial [bæk'tɪriəl] adj. 细菌的

viral ['vaɪrəl] adj. 病毒的

insect ['ɪnsekt] n. 昆虫, 虫子

pesticide ['pestisaɪd] n. 杀虫剂, 农药

microbial ecologist 微生物生态学家

nitrogen ['naɪtrədʒən] n. 氮

sulfur ['sʌlfə] n. 硫

bioremediation [baɪə'reɪmɪ:di'eɪʃən] n.

生物修复

Many important areas of microbiology do not deal directly with human health and disease but certainly contribute to human welfare. Agricultural microbiology is concerned with the impact of microorganisms on agriculture. Agricultural microbiologists try to combat plant diseases that attack important food crops, work on methods to increase soil fertility and crop yields, and study the role of microorganisms living in the digestive tracts of ruminants such as cattle. Currently there is great interest in using bacterial and viral insect pathogens as substitutes for chemical pesticides.

The field of microbial ecology is concerned with the relationships between microorganisms and their living and nonliving habitats. Microbial ecologists study the contributions of microorganisms to the carbon, nitrogen, and sulfur cycles in soil and in freshwater. The study of pollution effects on microorganisms is also important because of the impact these organisms have on the environment. Microbial ecologists are employing microorganisms in bioremediation to reduce pollution effects.

toxin ['tɒksɪn] n. 毒素

microbial energy 微生物能源

nitrogen fixation 固氮作用

microbial growth 微生物生长

survival [sə'vaɪvəl] n. 生存, 存活

microbial genetics 微生物遗传学

genetic information 遗传信息

microbial geneticist 微生物遗传学家

strain [streɪn] n. 菌株

hormone ['hɔ:məʊn] n. 荷尔蒙, 激素

Research on the biology of micro-organisms occupies the time of many microbiologists and also has practical applications. Those working in microbial physiology and biochemistry study the synthesis of antibiotics and toxins, microbial energy production, the ways in which microorganisms survive harsh environmental conditions, microbial nitrogen fixation, the effects of chemical and physical agents on microbial growth and survival, and many other topics.

Microbial genetics and molecular biology focus on the nature of genetic information and how it regulates the development and function of cells and organisms. The use of microorganisms has been very helpful in understanding gene function. Microbial geneticists play an important role in applied microbiology by producing new microbial strains that are more efficient in synthesizing useful products.

微生物遗传学和分子生物学主要研究的是遗传信息的本质以及它如何调节细胞以及生物体的发育和功能。利用微生物对我们理解基因的功能是非常有用的。微生物遗传学家通过制造更为有效地合成有用产物的微生物新菌株，对微生物学的应用起到了重要作用。

As it is known, microbiology has had a profound influence on society. What of the future? Science writer Bernard Dixon is very optimistic about microbiology's future for two reasons. First, microbiology has a clearer mission than do many other scientific disciplines. Second, it is confident of its value because of its practical significance. Dixon notes that microbiology is required both to face the threat of new and reemerging human infectious diseases and to develop industrial technologies that are more efficient and environmentally friendly.

**What are some of the most promising areas for future microbiological research and their potential practical impacts?
What kinds of challenges do microbiologists face?**

The following brief list should give some idea of what the future may hold:

1. New infectious diseases are continually arising and old diseases are once again becoming widespread and destructive. AIDS, hemorrhagic fevers, and tuberculosis are excellent examples of new and reemerging infectious diseases. Microbiologists will have to respond to these threats, many of them presently unknown.

2. Microbiologists must find ways to stop the spread of established infectious diseases. Increases in antibiotic resistance will be a continuing problem, particularly the spread of multiple drug resistance that can render a pathogen impervious to current medical treatment. Microbiologists have to create new drugs and find ways to slow or prevent the spread of drug resistance. New vaccines must be developed to protect against diseases such as AIDS. It will be necessary to use techniques in molecular biology and recombinant DNA technology to solve these problems.

微生物学家必须设法阻止已经被证实的传染病的传播。微生物的抗生素抗性的增强将是一个持久的问题，尤其是多种药物抗性的传播会使得当今医学治疗方法无法处置病原体。微生物学家不得不开发新的药物并找到减缓或阻止药物抗性传播的方法。新疫苗必须被开发出来以使人们免受像艾滋病等这样的疾病的威胁。我们必须使用微生物学新技术和重组DNA技术解决这些问题。

3. Research is needed on the association between infectious agents and chronic diseases such as autoimmune and cardiovascular diseases. It may be that some of these chronic afflictions partly result from infections.

4. We are only now beginning to understand how pathogens interact with host cells and the ways in which diseases arise. There also is much to learn about how the host resists pathogen invasions.

5. Microorganisms are increasingly important in industry and environmental control, and we must learn how to use them in a variety of new ways. For example, microorganisms can (a) serve as sources of high-quality food and other practical products such as enzymes for industrial applications, (b) degrade pollutants and toxic wastes, and (c) be used as vectors to treat diseases and enhance agricultural productivity. There also is a continuing need to protect food and crops from microbial damage.

6. Microbial diversity is another area requiring considerable research. Indeed, it is estimated that less than 1% of the earth's microbial population has been cultured. We must develop new isolation techniques and an adequate classification of microorganisms, one which includes those microbes that cannot be cultivated in the laboratory. Much work needs to be done on microorganisms living in extreme environments. The discovery of new microorganisms may well lead to further advances in industrial processes and enhanced environmental control.

7. Microbial communities often live in biofilms, and these biofilms are of profound importance in both medicine and microbial ecology. Research on biofilms is in its infancy; it will be many years before we more fully understand their nature and are able to use our knowledge in practical ways. In general, microbe-microbe interactions have not yet been extensively explored.

8. The genomes of many microorganisms already have been sequenced, and many more will be determined in the coming years. These sequences are ideal for learning how the genome is related to cell structure and what the minimum assortment of genes necessary for life is. Analysis of the genome and its activity will require continuing advances in the field of bioinformatics and the use of computers to investigate biological problems.

许多微生物基因组已经被测序，并且更多微生物基因组将在未来被鉴定。这些序列非常有助于我们了解基因组与细胞结构的相互关系以及什么是生命所需必须的最低限度。基因组及其活性的分析将需要人们在生物信息学领域的不断取得进展并将计算机应用在生物学问题的研究方面。

9. Further research on unusual microorganisms and microbial ecology will lead to a better understanding of the interactions between microorganisms and the inanimate world. Among other things, this understanding should enable us to more effectively control pollution. Similarly, it has become clear that microorganisms are essential partners with higher organisms in symbiotic relationships. Greater knowledge of symbiotic relationships can help improve our appreciation of the living world. It also will lead to improvements in the health of plants, livestock, and humans.

10. Because of their relative simplicity, microorganisms are excellent subjects for the study of a variety of fundamental questions in biology. For example, how do complex cellular structures develop and how do cells communicate with one another and respond to the environment?

11. Finally, microbiologists will be challenged to carefully assess the implications of new discoveries and technological developments. They will need to communicate a balanced view of both the positive and negative long-term impacts of these events on society.

The future of microbiology is bright. The microbiologist René Dubos has summarized well the excitement and promise of microbiology: How extraordinary that, all over the world, microbiologists are now involved in activities as different as the study of gene structure, the control of disease, and the industrial processes based on the phenomenal ability of microorganisms to decompose and synthesize complex organic molecules. Microbiology is one of the most rewarding of professions because it gives its practitioners the opportunity to be in contact with all the other natural sciences and thus to contribute in many different ways to the betterment of human life.